

29 Ans: C

$\Delta m c^2 = \text{Energy of photon emitted}$

$$(m_{\text{excited}} - 59.9308 \text{ u}) c^2 = 2.13 \times 10^{-13} \text{ J}$$

$$(m_{\text{excited}} - 59.9308 \text{ u}) = 2.366667 \times 10^{-30} \text{ kg}$$

Since $1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$, $(m_{\text{excited}} - 59.9308 \text{ u}) = 0.0014257 \text{ u}$

$$m_{\text{excited}} = 59.9322 \text{ u (C)}$$

30 Ans: C